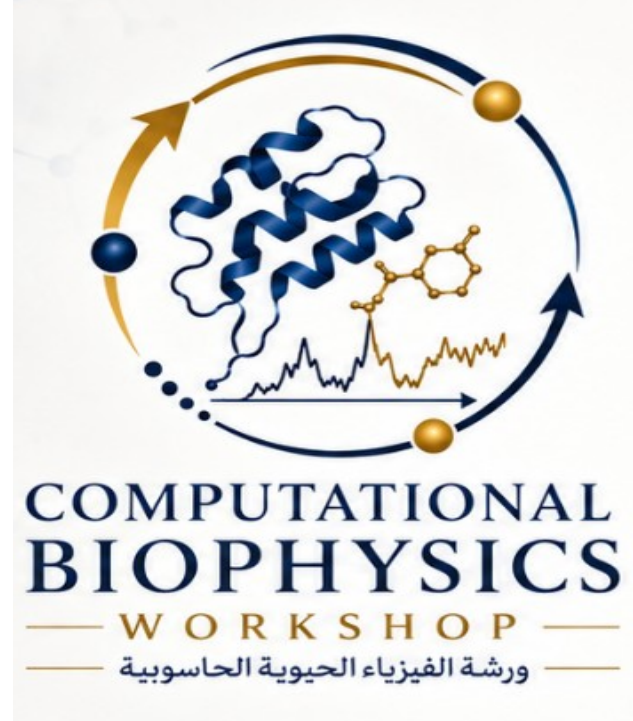


Molecular Docking — From **Theory** to **Practice**



Today's roadmap



Foundations

What docking is, virtual screening, the PDB file format, and the questions docking answers.



The theory with a game



Models & outputs

Three models of how molecules bind, and what a docking run actually returns.



Practical workflow

Preparing ligands and receptors, finding the active site, and running AutoDock Vina.

01

Foundations

What docking is, the screening problem it solves, and the file format that carries protein structures.

What is docking?

DEFINITION

Docking is the process of generating many possible 3D shapes (conformations) for a small molecule when it binds to a receptor, then ranking them by predicted binding strength.

WHY IT MATTERS

Instead of synthesizing thousands of molecules in a lab and testing each one, docking lets a computer screen millions of virtual compounds and predict which ones are most likely to bind.

KEY TERM

Virtual screening (VS) — running docking on large databases of small molecules (free or commercial) to search for drugs that bind to a macromolecule of interest.



Two flavors of virtual screening

Both aim to find new drug candidates, but they use different starting information.

01 Structure-based VS

WHEN TO USE

You have the 3D structure of the target protein (e.g. from X-ray, cryo-EM, or AlphaFold).

Docking software slides millions of small molecules into the protein's binding pocket, scores each fit, and ranks them.

The top-ranked compounds become candidates for synthesis and lab testing.

- Predicts binding pose and affinity
- Needs a 3D receptor structure
- Works best when the active site is known

EXAMPLE TOOLS

AutoDock Vina · GNINA · Glide

02 Ligand-based VS

WHEN TO USE

You know several active molecules (known binders) but have no 3D structure of the protein.

Builds a statistical or shape model from known active compounds, then searches large databases for molecules that look similar. The assumption: molecules similar to known actives are likely to act the same way.

- No receptor structure needed
- Uses known active compounds as templates
- Includes QSAR, pharmacophore, and similarity search

EXAMPLE TOOLS

RDKit · ROCS · Phase



This workshop focuses on **structure-based VS** — docking into a known 3D receptor.

Where does the protein structure come from?

Two routes give us the 3D coordinates of a receptor — experimental and theoretical. Both usually end up in a .pdb file.

01 Experimental techniques

- **X-ray crystallography** — beam a crystal of the protein with X-rays; the diffraction pattern reveals atom positions.
- **NMR spectroscopy** — measure magnetic responses of atomic nuclei in solution; works for smaller proteins.
- **Cryo-EM** — freeze the protein in vitreous ice and image it with an electron microscope; great for large complexes.

02 Theoretical techniques

- **Homology modeling** — build a 3D model from the sequence of a similar protein whose structure is known.
- **AlphaFold / AI prediction** — predict the structure from the amino-acid sequence alone using deep learning.

Protein Data Bank Format

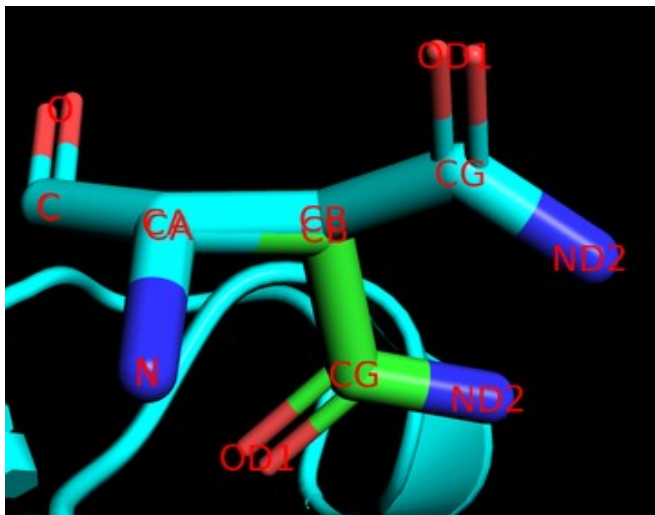
Record Type	Columns	Data	Justification	Data Type
ATOM	1-4	“ATOM”	left	character
	7-11	Atom serial number	right	integer
	13-16	Atom name	left*	character
	17	Alternate location indicator		character
	18-20	Residue name	right	character
	22	Chain identifier		character
	23-26	Residue sequence number	right	integer
	27	Code for insertions of residues		character
	31-38	X orthogonal Angstrom coordinate	right	floating
	39-46	Y orthogonal Angstrom coordinate	right	floating
	47-54	Z orthogonal Angstrom coordinate	right	floating
	55-60	Occupancy	right	floating
	61-66	Temperature factor	right	floating
	73-76	Segment identifier (optional)	left	character
	77-78	Element symbol	right	character
	79-80	Charge (optional)		character
HETATM	1-6 7-80	“HETATM” same as ATOM records		
TER	1-3	“TER”		character

Occupancy and amino-acid conformations

TWO KEY COLUMNS

Each ATOM record carries two columns that beginners often miss:

- **occupancy** — how often that atom appears at this position across the experiment (0.0 to 1.0)
- A single protein can have multiple **ATOM** records for the same atom — alternative conformations (**altLoc**) split between residue variants



CZ3	TRP	A	103	-2.718	-17.602	-7.823	1.00
CH2	TRP	A	103	-2.075	-18.703	-8.391	1.00
N	AASN	A	104	-5.450	-14.133	-13.066	0.46
N	BASN	A	104	-5.462	-14.119	-13.057	0.54
CA	AASN	A	104	-4.879	-14.137	-14.411	0.46
CA	BASN	A	104	-4.927	-14.115	-14.414	0.54
C	AASN	A	104	-4.173	-12.839	-14.796	0.46
C	BASN	A	104	-4.188	-12.835	-14.797	0.54
O	AASN	A	104	-3.716	-12.692	-15.926	0.46
O	BASN	A	104	-3.713	-12.702	-15.920	0.54
CB	AASN	A	104	-5.968	-14.444	-15.442	0.46
CB	BASN	A	104	-6.052	-14.377	-15.426	0.54
CG	AASN	A	104	-6.360	-15.909	-15.456	0.46
CG	BASN	A	104	-7.283	-13.506	-15.189	0.54
ND2	AASN	A	104	-7.659	-16.169	-15.525	0.46
ND2	BASN	A	104	-8.421	-13.953	-15.711	0.54
OD1	AASN	A	104	-5.507	-16.795	-15.398	0.46
OD1	BASN	A	104	-7.217	-12.455	-14.548	0.54

Why does it matter for docking?

Proteins are not static rocks — the same residue can occupy two or more slightly different positions in the same crystal.

- ⚙️ A residue near the binding pocket may have two rotamer positions:
 - pick the wrong one and your docked ligand may not fit.
 - Keep both of them and you will get an excellent affinity. is this correct?
 - ⚠️ Always inspect **occupancy** and **altLoc** columns when preparing a receptor — delete low-occupancy atoms.
-
- 🔧 Tools like **PyMOL** and **AutoDock Tools** let you clean these alternatives before docking.

02

The Theory Behind Docking

How a computer searches through millions of possible ligand positions, scores them, and validates that the answer is trustworthy.

What happens **behind the scenes**?

Every docking program runs **two engines in series**: a search engine that generates poses, and a scoring engine that ranks them.

1 Step 1 — Search algorithm

GENERATES POSES

A conformational search algorithm samples **all degrees of freedom** of the ligand — translational (where in the pocket), rotational (which way it points), and conformational (which rotatable bonds are twisted).

The goal: **generate a diverse set of candidate poses**, not just one.



Genetic algorithm



Monte Carlo



2 Step 2 — Scoring function

RANKS POSES

For every generated pose, a scoring function estimates how strongly the ligand binds. The score is a number — **lower (or higher, depending on program)** means stronger binding.

▽ Knowledge-based — learned from statistical patterns in known protein-ligand complexes

◆ Force-field based — physics equations for electrostatics, van der Waals, bond energies

◆ Empirical — fitted formula:
Energy $\approx W1 \cdot \text{H-bond} + W2 \cdot \text{hydrophobic} + W3 \cdot \text{electrostatic}$

Scoring the generated poses

No scoring function is perfect — every choice is a **trade-off between accuracy and speed**.

THE TRADE-OFF

There are different techniques for predicting binding free energies from 3D structures of a protein-ligand complex. Each one balances accuracy and speed differently. The right choice depends on how many molecules you need to screen and how precise you need to be.

TWO ENDS OF THE SPECTRUM

- ⚡ Fast scoring functions — process thousands of compounds per hour, ideal for virtual screening. The trade-off: they may miss subtle binding effects and produce false positives.
- Accurate scoring functions — Free Energy Perturbation (FEP), Thermodynamic Integration (TI). They give excellent binding energy estimates but can take hours per compound.

IN PRACTICE

Most docking campaigns use a fast scoring function for the initial screen, then re-rank the top 100 hits with a more accurate method before synthesis. This is sometimes called consensus scoring.

Redocking — **validate before you trust** a scoring function

Before running VS on thousands of compounds, check whether your docking program can **reproduce a known answer**.

THE PROCEDURE

- 1- Take a protein-ligand complex from the PDB where the ligand is already bound.
- 2- Remove the ligand.
- 3- Re-dock it back into the same pocket using the scoring function you plan to use.
- 4- Then compare the predicted pose to the experimentally observed pose.

THE METRIC — RMSD

RMSD (Root Mean Square Deviation) measures the average distance, in Ångströms, between the atoms of the predicted pose and the experimental pose. Lower is better.

THE THRESHOLD

If **RMSD < 2.0 Å**, the predicted pose is considered near-native — the scoring function is reliable enough for virtual screening with this protein. Above 2.0 Å, look for another scoring function or tweak the protocol.

03

Models of Binding & Docking Output

Three classic models of how molecules recognize each other, and what a docking run actually hands back to you.

What is the **result of docking**?

A docking run gives you **three things** — together they answer: how does the molecule bind, and how strongly?



01

Ligand conformation

The 3D shape the ligand adopts when bound — its rotatable bonds settled into the conformation the algorithm predicts is most favorable inside the pocket.



02

Relative orientation

Where in the pocket the ligand sits and which way it points — translation, rotation, and the specific residues it interacts with (H-bonds, hydrophobic contacts).



03

Binding score

A number estimating the goodness of the interaction — typically a binding free energy (kcal/mol) or a unitless score. Usually lower energy = stronger predicted binding.



IN PRACTICE

Docking produces a ranked list of poses for each ligand. You then visually inspect the top 5–10 poses in PyMOL before deciding which compounds to use for next step.

Three models of binding

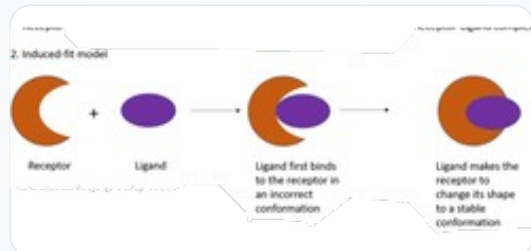
How does a small molecule recognize its protein target? Three competing models explain the mechanism — and inform how docking software treats **receptor flexibility**.



01 Lock and Key

The receptor is treated as rigid — a fixed lock. Binding happens only when the ligand's shape is geometrically complementary to the binding site, like a key in a lock.

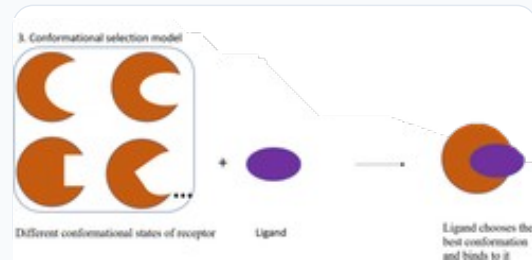
Proposed by Emil Fischer in **1894**; it is fast but ignores flexibility.



02 Induced Fit

The ligand binds first, then the protein adjusts — side chains and backbone move to wrap around the ligand, like a glove shaping to a hand.

Daniel Koshland's **1958** model explains why some substrates bind better than their rigid shape would suggest.



03 Conformational Selection

The protein constantly samples multiple conformations even without ligand. The ligand selectively binds to one of these pre-existing shapes, shifting the equilibrium. This is the modern view; it implies docking should consider receptor ensembles.

① Most docking programs still assume the lock-and-key model for speed. Induced-fit docking and ensemble docking are slower but more realistic.

04

Practical Workflow

What you actually need to run a docking job, and the step-by-step protocol with AutoDock Vina and OpenBabel.

What do we need to perform docking?

Four ingredients.

01



Prepared receptor

A 3D protein structure in .pdbqt format, with waters removed, polar hydrogens added.

02



Prepared ligands (3D)

Each ligand as a separate .pdbqt file with 3D coordinates, protonation states assigned, and rotatable bonds marked. Docking needs 3D — 2D or 1D (e.g. SMILES) is not enough.

? What if 2D or 1D? → convert with OpenBabel or CDPKit

03



Known active site

The (x, y, z) coordinates of the binding pocket center, plus a box size that defines where the ligand is allowed to search.

? What if not known? → predict with P2Rank, CASTp, DeepSite, Fpocket, etc...

04



Docking program

Software that runs the search + scoring loop. AutoDock Vina and GNINA are free and widely used; commercial alternatives exist (Schrödinger Glide, Gold).

How to know the active site?

If your protein has no known ligand, you need to predict where the binding pocket is.

← Strategy 1 — Sequence alignment

Find a similar protein (with the same biological function) from a different species whose active site IS known. Align the two sequences. Residues that are conserved across species are likely important for function — they often cluster around the active site.

⊙ Strategy 2 — Computational prediction

Run a pocket-detection program on your 3D structure.

◆ **P2Rank** — fast, standalone Java tool, GitHub

○ **DeepSite** — deep-learning-based, web service

☁ **CASTp** — web server, computes pocket volume and area

◆ **FPocketWEB** — detects alpha-sphere-based pockets

Programs & resources

Everything you need to start docking — all free, all open source.



AutoDock Vina (v1.2.7)

The docking engine itself — fast, open-source, command-line driven.

<https://github.com/ccsb-scripps/AutoDock-Vina/releases/tag/v1.2.7>



GNINA (v1.3.3)

Docking with optional CNN-based rescoring — often more accurate than Vina alone.

<https://github.com/gnina/gnina>



AutoDock Tools (MGLTools)

GUI for preparing receptor .pdbqt files and defining the grid box.

<https://ccsb.scripps.edu/mgltools/downloads/>



PyMOL

Visualize and clean PDB structures before docking.

<https://www.pymol.org/>



OpenBabel

Convert formats, generate 3D, minimize, compute RMSD.

<https://github.com/openbabel/openbabel/releases>



Google Colab notebook

Run GNINA docking in the cloud — no local install needed. Pre-built GNINA Colab available. (at the end of the WS)

Ligand preparation — step by step

01 Obtain ligands in 3D (.pdbqt format)

Download the compounds SMILES from a database (ZINC, ChEMBL, DrugBank). Then convert using obabel to .pdbqt format (AutoDock Vina needs .pdbqt — a PDB file with charge and atom-type columns added.).

Each ligand in its own file.

protonate with `dimorphite_dl` and rename using `rename_ph.py`

```
obabel *.smiles -O comp_*.pdbqt --gen3D --minimize --ff UFF --steps 100000
```

```
obabel *.smiles -O .pdbqt --gen3D --minimize --ff UFF --steps 100000 --split title (rename with compound title)
```

OR

```
obabel *.sdf -O *.pdbqt -p 7.4 (from sdf to pdbqt)
```

02 Put all .pdbqt files in one folder

Create a working folder containing all ligand .pdbqt files, plus the `conf.txt` (**I sent it**) and `vina.exe` (**you downloaded it**).

Create another subfolder inside the main folder and name it **rec** that will contain the prepared receptor (**next slide**).

⚡ USEFUL OPENBABEL ONE-LINER — SPLIT ONE FILE INTO INDIVIDUAL COMPOUND FILES:

```
obabel compounds.sdf -O comp_.sdf -m (rename sequentially)
```

```
obabel compounds.sdf -O .sdf --split title (rename with compound title)
```

Receptor preparation — step by step

01 Clean the receptor in PyMOL

Remove waters, unwanted chains, ligands, and low-occupancy atoms. Save as a clean .pdb file.

02 Convert .pdb → .pdbqt in AutoDock Tools

Open the .pdb in AutoDock Tools (ADT).

1- Edit → Hydrogens → Add → select polar only → click Ok

2- Grid → Macromolecule → choose → select the protein → click select molecule → then save as .pdbqt in the **rec** folder you created.

3- in the conf.txt write rec/###.pdbqt after receptor= (### in the name of the prepared protein file you saved)

03 Define the grid box

First method (if you know the numbers of amino acids):

1- open the protein-ligand complex in PyMol → select the ligand → click on A of the sele object → modify → around → residues within 4 angstroms

2- run this command in pymol to get the numbers of the amino acids surrounding the ligand and write them in a txt file:

iterate sele and name CA, print(resi, end=',')

3- in ADT, search for these numbers and show them as spheres → continue from the **Grid step below** (Grid → Grid Box ...).

Second Method (if you have the cocrystallized ligand):

1- Open the cocrystallized ligand in ADT and show it as spheres.

2- Grid → Grid Box → **change spacing to 1.000** → change the center and size so that the box enclose the cocrystallized ligand/amino acids.

3- In the conf.txt file write the values of the center and size.

Performing docking

- 01 Run docking**
From the working folder (containing **vina.exe**, all ligand **.pdbqt** files, **conf.txt**, and the **rec/ subfolder**), run the docking loop:
for %f in (*.pdbqt) do vina_1.2.7_win.exe --seed 5 --ligand "%f" --config conf.txt > "%~nf.log"
- 02 Summarize affinities with [summarize_logs.bat](#) (I will send it). Click on it to run**
- 03 validate redocking RMSD with this command:**

obrms REF.sdf OUT.pdbqt -f

REF: is the reference ligand file
OUT: is the output poses of the cocrystalized ligand